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# Supplementary Material: Repair-Aware Quantum Sampling for a Discrete Process-Design QUBO

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## ABSTRACT

This supplement records provenance, validation, and artifact anchors for the DS-MFG repair-aware quantum-sampling manuscript. It is intended to be compiled and submitted separately from the main conference paper if appendices are not counted as article body material in the submission workflow.

Keywords: supplementary material, quantum optimization, process design, reproducibility

## REPOSITORY AND PACKAGING

The main manuscript keeps only the narrative-critical values. This supplement keeps the audit data close to the manuscript source while allowing the main article to remain within the conference page limit. The repository URL is https://github.com/SECQUOIA/pse-quantum-discrete-optimization. A final archival DOI has not yet been assigned. It will be added before camera-ready submission if one is minted; the review artifact is identified by the repository snapshot that contains this supplement and by the checksums below.

Table 1: Current reproducibility pins validated for the DS-MFG case study.

| Layer | Current pin or check | Repository anchor |
|---|---|---|
| Julia optimization wrapper | `QiskitOpt.jl` v0.7.0; manifest git tree `a601dccdd43fd0a8b18e63f3b771273817af49eb`. | `case-studies/ds-mfg-qubo-qiskitopt/Project.toml`; `case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml`. |
| Julia QUBO stack | `QUBODrivers.jl` v0.6.1, `QUBOTools.jl` v0.13.0, `QUBO.jl` v0.6.0. | `case-studies/ds-mfg-qubo-qiskitopt/Project.toml`; `case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml`. |
| Python bridge | Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, Qiskit IBM Runtime 0.46.1, NumPy 2.2.6, SciPy 1.15.3. | `case-studies/ds-mfg-qubo-qiskitopt/CondaPkg.toml`. |
| Fresh-clone validation | `julia --project=. scripts/smoke_test.jl` imports packages, checks required artifacts, and validates cached summary values. | `case-studies/ds-mfg-qubo-qiskitopt/scripts/smoke_test.jl`. |

Table 2: Reproducibility levels used for the DS-MFG manuscript package.

| Level | Meaning | Current support |
|---|---|---|
| Score-from-cache reproducible | Retained samples, summaries, and repair tables can be inspected and rescored without rerunning the stochastic campaigns. | Supported by cached distributions, hit-rate reports, checksums, and the smoke test. This is the main reproducibility level for the manuscript claims. |
| Rerunnable with retained scripts | The scripts and pinned environment can rerun selected QAOA/VQE, classical, and hardware-handoff workflows, subject to stochastic variation and credential availability. | Partially supported. QAOA fixed-angle parameters and sampled outputs are retained; selected VQE metadata reruns now retain optimized ansatz vectors for seed 74018 and for the final-budget selected seeds 74018, 74007, and 74001. |
| Full campaign replay | Every stochastic campaign can be reproduced byte-for-byte from retained optimizer states, simulator logs, hardware calibration, and submitted-circuit metadata. | Not claimed. Some optimizer, MPS truncation, and submitted-job hardware metadata were not retained. |

Table 3: Checksums for retained archives and primary cached summaries.

| Artifact | SHA-256 checksum |
|---|---|
| `case-studies/ds-mfg-qubo-qiskitopt/Fw_DSmfgcasequboinformation.zip` | `aa20d793fc45c25c2bd7cc0a231877f71b225bf80daee11531cef6eb6cac5af2` |
| `ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip` | `88af6f74b0fc10dcece69e783c37782649d1082e3220a88c61f389274665dff9` |
| `ds_mfg_direct_full_qubo_hardware_pilot.zip` | `fe7a60bd6576e66da9059d8aa6831424ef6fcbbf47bd934e60850b7933558091` |
| `case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_qaoa_juliqaoa_transfer_highread/qaoa_juliqaoa_transfer_summary.csv` | `10bb3bc9b746fb12a87eadc1fd2b819c1a3cacbc84025c9bd54797c1088c34b7` |
| `case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_final/vqe_reduced_top50_sampling_summary.csv` | `aac0299d49314b8d0e4d54b088f50cc728f1d212540613348ab62d209a132ee1` |
| `case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/ip_exact_enumeration_summary.json` | `1298dcb07aa82c0adc05ead3fb1b3f4b86ae9ef38e9a276b98f97ec5a7606e5e` |
| `case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json` | `13f7683c41cbf14b4225326e41778c6f42ae3979e4d1303af65c7132fe35a6e7` |
| `case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_selected_metadata/vqe_reduced_top50_sampling_summary.csv` | `df82de49b921e3b70d5d24f0896e77b6da64477c3a13cffc0a0761d6e9ebd2db` |
| `case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_hit_rate_reports/time_to_solution_report.csv` | `c27e70684f3b753efc4eb8cf2b75e79afe4b15402bcd6b3ca7d7a598310f2efd` |

## VALIDATION FACTS

Table 4: Numerical validation facts used to support the manuscript claims.

| Check | Retained value | Anchor |
|---|---|---|
| Exact repaired optimum | $F(y^\star) = 11.7095$ at flow 1001110100111100011. | `ds_mfg_reduced_flow_objective/reduced_flow_summary.csv`; `ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv`. |
| Gurobi pool bridge | All 36 stored Gurobi-pool flows match the repaired QUBO score within $3.1 \times 10^{-13}$. | `MANUSCRIPT_FINDINGS.tex`; `Fw_DSmfgcasequboinformation.zip`. |
| Retained Gurobi metadata | Model capex: 19 binary variables, 20 constraints, 58 nonzeros; `PoolSearchMode=2`, `PoolSolutions=36`, solve time 0.073 s, 36 solutions found. Solver version, status code, MIPGap, PoolGap, and pool completeness/diversity intent were not retained in the original export. A local Gurobi 13.0.2 rerun with `PoolSearchMode=2` and `PoolSolutions=100` returned `OPTIMAL`, `MIPGap=0.0`, 36 solutions, and exact retained-pool agreement. | `Fw_DSmfgcasequboinformation.zip/gurobi_pool_solutions_n36_20260126_105706.json`. |
| Quadratic surrogate weakness | Global fit $R^2 \approx 0.999$, but the surrogate optimum repairs to 14.5505. The true repaired optimum is surrogate rank 33; the surrogate optimum is exact repaired rank 6. | `ds_mfg_reduced_flow_objective/reduced_flow_summary.csv`; Figure 1. |
| Low-energy surrogate audit | The exact top-50 repaired flows include 14 non-pool flows. In the lowest 1% repaired-flow set, exact-rank versus surrogate-rank Spearman correlation is 0.901. Exact enumeration wall time was not retained. | `ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv`; `MANUSCRIPT_FINDINGS.tex`. |
| Primary QAOA simulator check | JuliQAOA statevector probability on the reference optimum was $2.63 \times 10^{-3}$ for the energy-targeted $p = 5$ angles. The cached Aer MPS sampled rate was $2.53 \times 10^{-3}$, inside its Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$. MPS truncation logs were not retained. | `ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv`; `ds_mfg_qaoa_juliqaoa_transfer_highread/qaoa_juliqaoa_transfer_summary.csv`. |

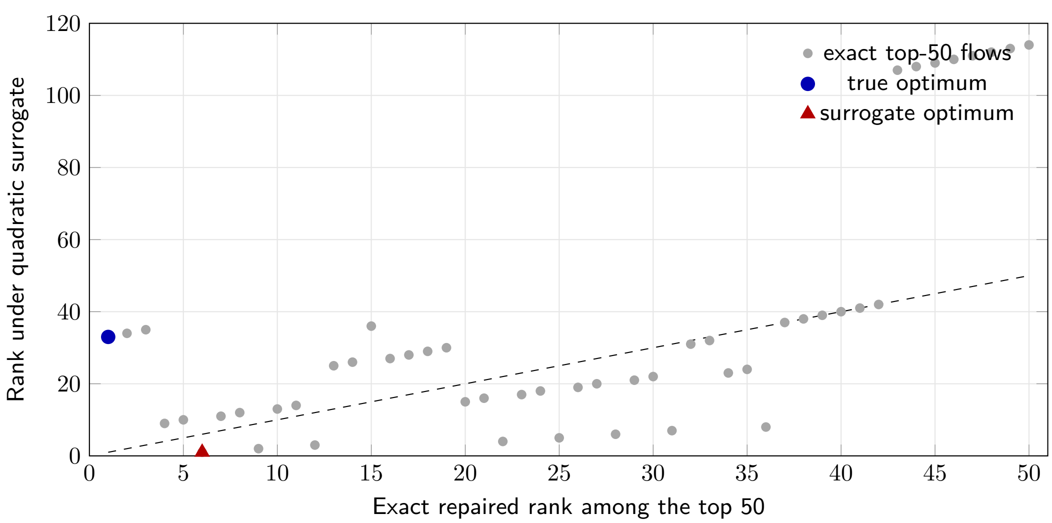


Figure 1. Surrogate rank audit for the exact top-50 repaired flows. The quadratic surrogate has high global $R^2$ but imperfect low-energy ordering: the exact optimum is only rank 33 under the surrogate, while the surrogate minimum repairs to the sixth-best exact flow. Quantum-sampling outcomes are therefore scored using the exact repaired objective $F$, not the surrogate $\widehat{F}$.

Table 5: Implications of missing provenance values for the manuscript claims.

| Missing value | Affects main claim? | Affects full rerun? | Current mitigation |
|---|---|---|---|
| Historical Gurobi final status and MIPGap | No for the current claim after rerun; the original export still lacks these fields, but the model/pool behavior has been rerun locally. | Partly; exact historical solver state is unavailable, but current solver provenance is retained. | The same flow is certified by exact enumeration over all $2^{19}$ flows and by a local Gurobi 13.0.2 rerun with `OPTIMAL` status, `MIPGap=0.0`, 36 solutions, and exact retained-pool agreement. |
| MPS truncation logs | No for the central QAOA sampling claim. | Yes, for simulator-forensics replay. | The energy-targeted QAOA row is cross-checked against independent JuliQAOA statevector probabilities, which agree with the Aer MPS sampled rate within the Wilson interval. |
| VQE optimized parameter vectors | No; VQE is secondary and reported as a retained sampled distribution. | Partly; selected fixed-circuit replay is now supported, but full historical optimizer replay is not claimed. | Metadata reruns retain optimized `EfficientSU2` vectors for seed 74018 and for seeds 74018, 74007, and 74001 at the final sampling budget. The current metadata rerun produced 4 reference hits total and is provenance, not a replacement for the historical 28-hit final follow-up. |
| Submitted hardware layout, depth, and two-qubit count | No, because the hardware row is a submission-and-scoring handoff only. | Yes, for hardware-performance diagnosis. | The manifest retains backend, job IDs, shots, seeds, logical circuit metadata, and scored counts. No noise-mechanism claim is made. |
| Hardware calibration or mitigation metadata | No under the current claim. | Yes, for calibrated hardware analysis. | The manuscript states that no readout or error mitigation is claimed and that the retained metadata cannot diagnose a specific noise source. |
| Exact enumeration wall time | No. | Partly, for cost accounting. | The enumeration is documented as exact over all $2^{19}$ flows; issue 24 tracks recording wall time if the enumeration script is rerun. |

## VARIATIONAL AND HARDWARE PROVENANCE

Table 6: Variational-parameter provenance for the central QAOA and VQE rows.

| Workflow | Circuit / ansatz | Objective and optimizer | Parameter source | Validation metric |
|---|---|---|---|---|
| Energy-targeted QAOA | $p = 5$ QAOA on the 19-flow surrogate. | Minimize $\mathbb{E}_{p_\theta}[\widehat{F}(y)]$ with JuliQAOA `find_angles_bh`; seed 91001; 5 basin-hopping iterations. | `ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv`; angles transferred beta-then-gamma to `QiskitOpt.QAOA`. | 664 reference hits / 262144 reads; 39661 top-50 reads. |
| Top-10 QAOA diagnostic | $p = 5$ QAOA on the 19-flow surrogate. | Maximize probability on the enumerated exact top-10 set; oracle-aided because the repaired ranking is known. | Top-10 angle vector retained in `ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip/job_manifest.json`; local 1007-hit diagnostic CSV was not retained. | 1007 reference hits / 262144 reads in the retained manuscript summary; hardware handoff used this circuit. |
| Reduced-surrogate VQE | `EfficientSU2`; 152 initial parameters on the 19-flow surrogate. | COBYLA through `QiskitOpt.VQE`; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 in the final follow-up. | Historical sampled distributions retained; current metadata reruns persist optimized ansatz vectors in `ds_mfg_vqe_reduced_flow_objective_seed74018_metadata/` and `ds_mfg_vqe_reduced_flow_objective_selected_metadata/`. | Historical final follow-up: seed 74018 had 20 reference hits / 524288 reads; three seeds had 28 reference hits / 1572864 reads. Current metadata rerun: 4 reference hits / 1572864 reads. |

Table 7: Hardware pilot provenance retained in the submitted-job archive.

| Item | Retained metadata | Limitation |
|---|---|---|
| Backend and jobs | `ibm_fez`; backend queried 2026-06-17 20:19 UTC; 156 qubits; 9 completed jobs; 4096 shots per job; repeats 1–3; transpile seeds 92001–92003; job IDs retained. | Queue time, calibration snapshot, readout mitigation, and error mitigation metadata were not retained. |
| Logical circuit | 19 qubits and 19 classical bits; pre-transpile depth 86; operation counts: 19 `h`, 95 `rx`, 95 `rz`, 255 `rzz`, and 19 measurements; scoring reverses Qiskit count keys to $x_1, \ldots, x_{19}$. | Submitted-job transpiled depth, two-qubit-gate count, physical layout, and measurement map were not retained. |
| Hardware outcome | 36864 total reads; 6 top-50 reads; 1 top-10 read; 4 stored Gurobi-pool reads; 0 reference-optimum reads; best repaired flow rank 2 with objective 11.8105. | The pilot validates submission and scoring only; it cannot diagnose a specific noise mechanism. |